This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scripter Modules, refer to the "Guide to Using Scripter Modules" document.

Device Specifications

Device Type: Other
Manufacturer: Ametek
Firmware Version: 2.04.281
Model(s): Axess Elite SX-AX20E, Axess Elite SX-AX15E

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scripter Version
2.08.0002-b001	1.4.2

Version History

Module Version	Date	Notes
1_0_0_0	8/14/2018	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
  Example: `InterfaceName.Unidirectional = 'True'`

- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
  Example: `InterfaceName.connectionCounter = 5`

- If login credentials are required, devicePassword and deviceUsername must be set accordingly. devicePassword and deviceUsername default values are admin and admin respectively.
  Example:
  ```
  InterfaceName.deviceUsername = 'admin'
  InterfaceName.devicePassword = 'admin'
  ```

Supported Classes and Examples

SerialClass

```
InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='Axess Elite SX-AX20E')
```

SerialOverEthernetClass

```
InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='Axess Elite SX-AX20E')
```

EthernetClass

```
InterfaceName = ModuleName.EthernetClass('192.168.254.254', 23, Model='Axess Elite SX-AX20E')
```

Control Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command	Value	Value	Value
OutletControl	'On'	'Off'	'Reboot'
Qualifier Key	Qualifier Value		
'Outlet'	'1' – '8'		

```
# OutletControl example
InterfaceName.Set('OutletControl', 'On', {'Outlet': '1'})
```

Command	Value	Value	Value
RelayControl	'On'	'Off'	'Reboot'
Qualifier Key	Qualifier Value	Qualifier Value	
'Relay'	'Aux 1'	'Aux 2'	

```
# RelayControl example
InterfaceName.Set('RelayControl', 'On', {'Relay': 'Aux 1'})
```

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
```

FeedbackHandler will be called only when the specified qualifier gets a new status.

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
```

FeedbackHandler will be called when any qualifier gets a new status.

Command	Value	Value	Value
BaudRate	'2400'	'9600'	'57600'
	'115200'		

```
# BaudRate examples
InterfaceName.Update('BaudRate')
Value = InterfaceName.ReadStatus('BaudRate')
InterfaceName.SubscribeStatus('BaudRate', None, FeedbackHandler)
```

Command	Value	Value
ConnectionStatus	'Connected'	'Disconnected'

```
# ConnectionStatus examples
Value = InterfaceName.ReadStatus('ConnectionStatus')
InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)
```

Command	Value	Value
OutletControl	'On'	'Off'
Qualifier Key	Qualifier Value	
'Outlet'	'1' – '8'	

```
# OutletControl examples
InterfaceName.Update('OutletControl', {'Outlet': '1'})
Value = InterfaceName.ReadStatus('OutletControl', {'Outlet': '1'})
InterfaceName.SubscribeStatus('OutletControl', None, FeedbackHandler)
```

Command	Value	Value
RelayControl	'On'	'Off'
Qualifier Key	Qualifier Value	Qualifier Value
'Relay'	'Aux 1'	'Aux 2'

```
# RelayControl examples
InterfaceName.Update('RelayControl', {'Relay': 'Aux 1'})
Value = InterfaceName.ReadStatus('RelayControl', {'Relay': 'Aux 1'})
InterfaceName.SubscribeStatus('RelayControl', None, FeedbackHandler)
```

Cable and Adapter Requirements

Captive Screw to Male DB9 RS-232 Serial Cable.

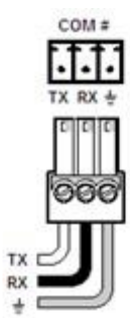
Notes for the Device

In order for the driver to work, Serial Port Requires Login must be **DISABLED**. To locate this setting, go to the device's embedded webpage under the Setup tab, go to Device.

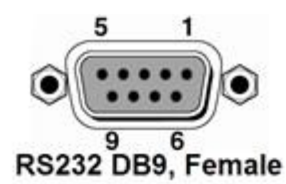
Serial communication

Port Type: RS-232
Baud Rate: 9600
Data Bits: 8
Parity: None
Stop Bits: One
Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD	⟶	2	RxD
RxD	⟵	3	TxD
GND	——	5	GND



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scripter ethernet interface

Port Type:	Ethernet
Default Port:	23
Logon Credentials Supported:	Yes
Default Username:	admin
Default Password:	admin
Multi-Connection Capabilities:	Undetermined
Port Changeability:	Yes

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

Appendix A. Set Commands

Outlet Control Off Outlet 1	`set outlet 1 off\x0D\x0A`
Outlet Control Off Outlet 8	`set outlet 8 off\x0D\x0A`
Outlet Control On Outlet 1	`set outlet 1 on\x0D\x0A`
Outlet Control On Outlet 8	`set outlet 8 on\x0D\x0A`
Outlet Control Reboot Outlet 1	`set outlet 1 reboot\x0D\x0A`
Outlet Control Reboot Outlet 8	`set outlet 8 reboot\x0D\x0A`
Relay Control Off Relay Aux 1	`set outlet 9 off\x0D\x0A`
Relay Control Off Relay Aux 2	`set outlet 10 off\x0D\x0A`
Relay Control On Relay Aux 1	`set outlet 9 on\x0D\x0A`
Relay Control On Relay Aux 2	`set outlet 10 on\x0D\x0A`
Relay Control Reboot Relay Aux 1	`set outlet 9 reboot\x0D\x0A`
Relay Control Reboot Relay Aux 2	`set outlet 10 reboot\x0D\x0A`

Appendix B. Update Commands

Baud Rate	`get console\x0D\x0A`
Outlet Control Outlet 1	`get outlet 1\x0D\x0A`
Outlet Control Outlet 8	`get outlet 8\x0D\x0A`
Relay Control Relay Aux 1	`get outlet 9\x0D\x0A`
Relay Control Relay Aux 2	`get outlet 10\x0D\x0A`